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# Step 1: Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Importing Linear Regression algorithm
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

# Step 2: Create dataset (PM2.5, PM10 → AQI)
data = {
    'PM25': [50, 80, 100, 120, 150, 180, 200, 220, 250, 300],
    'PM10': [80, 120, 150, 180, 200, 220, 250, 270, 300, 350],
    'AQI': [60, 90, 120, 150, 180, 210, 240, 270, 300, 350]
}

df = pd.DataFrame(data)

# Step 3: Define input (X) and output (Y)
X = df[['PM25', 'PM10']]
Y = df['AQI']

# Step 4: Split dataset
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=0
)

# Step 5: Apply Linear Regression Algorithm
# (This is the main algorithm used in the project)
model = LinearRegression()

# Step 6: Train model
model.fit(X_train, Y_train)

# Step 7: Predict AQI
Y_pred = model.predict(X_test)

# Step 8: Show results
print("Algorithm Used: Linear Regression")
print("Actual AQI:", Y_test.values)
print("Predicted AQI:", Y_pred)

# Step 9: Graph (PM2.5 vs AQI)
plt.scatter(df['PM25'], df['AQI'])
plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.title("AQI Prediction using Linear Regression")
plt.show()
```

Algorithm Used: Linear Regression

Actual AQI: [120 300]

Predicted AQI: [123.01206287 296.58949677]

